Till Aust

Konstanz, Germany | till.aust@uni-konstanz.de | linkedin.com/in/till-aust | Google Scholar

Machine Learning Researcher | PhD Candidate in Computer Science

Machine Learning for Physiological and Sensor Time Series in Cyber-Physical Systems | Collective and Multi-Robot Systems | 10+ peer-reviewed publications

Research interests: automated machine learning for time series, reliability and uncertainty of predictions under distribution shift, embedded machine learning on resource-constrained sensor nodes, physiological and plant electrophysiological signals, collective behavior and decentralized coordination in multi-robot systems.

Education

PhD Candidate in Computer Science (Dr.-Ing.) Nov 2022 – Present

Cyber-Physical Systems Group, University of Konstanz, Konstanz, Germany

Dissertation: Automated Machine Learning for Physiological Time Series Data in Cyber-physical Systems. Cumulative dissertation based on four peer-reviewed publications. Submission planned for November 2026.
Advisor: Prof. Heiko Hamann. Second referee: Prof. Felix Mohr.

Additional coursework: master's course Portfolio Management, Chair of Finance, University of Konstanz (winter term 2024/2025).

M.Sc. in Robotics and Autonomous Systems (Grade: 1.0, best possible) Oct 2019 – Jul 2022

University of Lübeck, Lübeck, Germany

Master's thesis at IRIDIA, Université Libre de Bruxelles, Belgium (Nov 2021 – Jul 2022): Collective Adaptation and Symmetry Breaking to Environmental Changes in a Swarm of Kilobots Monitoring a Dynamic Environment. Published at ANTS 2022.
Ran the physics-based simulation campaign in ARGoS on a High Performance Computing cluster with SLURM, and demonstrated transferability on 50 physical Kilobots on the Kilogrid.

Erasmus Exchange Semester Aug 2020 – Dec 2020

University of Bergen, Bergen, Norway

Focus on multi-agent systems and formal logic.

B.Sc. in Robotics and Autonomous Systems (Grade: 1.5, where 1.0 is best) Oct 2016 – Dec 2019

University of Lübeck, Lübeck, Germany

Bachelor's thesis on visual servoing for semi-automated 2D ultrasound scanning of peripheral arteries. Implemented the image-based detection and tracking of the artery and the control of a KUKA LBR iiwa robot arm. Published at AUTOMED 2020.

Research Experience

Visiting Researcher (Doctoral Research Stay) Apr 2026 – Jul 2026

National Institute of Informatics, Tokyo, Japan

Evaluated pretrained time-series foundation models in zero-shot inference on financial time series and developed reliability estimation for their forecasts under temporal distribution shift.
Designed a selective-classification layer that abstains from unreliable forecasts using data-efficient tabular classifiers, and assessed it with calibration analysis and risk-coverage curves.
Benchmarked the approach on regime-aware synthetic data under controlled distribution shifts, using a modular Python pipeline that runs thousands of parameterized evaluation jobs.

Research Assistant (Doctoral Researcher) Nov 2022 – Aug 2026

Cyber-Physical Systems Group, University of Konstanz, Konstanz, Germany

Conducted doctoral research on machine learning for physiological and sensor time series in the following projects:

WatchPlant (EU Horizon 2020 FET) 2022 – 2024

Applied automated machine learning and developed deep-learning classifiers for plant electrophysiological signals that turn living plants into biohybrid environmental sensors, detecting heat and ozone exposure with up to 0.98 sensitivity.
Deployed the classifiers on low-power embedded sensor nodes with ExecuTorch, co-developed the on-device C/C++ code, and rebuilt an operation missing from the runtime out of existing primitives.
Operated the sensor nodes in long-term outdoor field deployments and analyzed the resulting continuous, noisy measurement streams.

ChronoPilot (EU Horizon 2020 FET) 2022 – 2025
Estimated mental states and subjective time perception from time-synchronized wearable signals (electrocardiography and heart rate, electrodermal activity, eye tracking) for an adaptive human-machine feedback system.
Helped design and run the user studies, tested whether physiological features differ between task conditions, and validated the models across subjects on unseen participants. Assisted in further human-robot interaction studies.

AI4PlantPhysio_CO (German Federal Ministry of Education and Research) 2024 – 2025
Co-organized an international workshop and communicated the research to academic, industrial, and governmental stakeholders.

Conducted research on collective behavior in multi-agent systems:

Decentralized Synchronization (published at SAB 2026) 2025 – 2026
Studied how agents with firefly-inspired coupling reach collective timing without a central clock, and showed that noise and sparse connectivity can improve synchronization rather than degrade it.

Collaborated with electrical engineers, biologists, and psychologists in two EU consortia and translated their research questions into well-posed machine-learning problems.

Parallelized large parameter sweeps on a High Performance Computing cluster and containerized experiment pipelines with Docker and Kubernetes.

Research Assistant Aug 2022 – Nov 2022
Institute of Computer Engineering, University of Lübeck, Lübeck, Germany

Began doctoral research and moved with the research group to the University of Konstanz.

Student Research Assistant 2020
Institute of Computer Engineering, University of Lübeck, Lübeck, Germany

Conducted simulation-based research on robot swarms and contributed to a peer-reviewed publication on collaboration and interference in robot swarms (ANTS 2020).

Teaching and Supervision

Supervision and Teaching Nov 2022 – Aug 2026
Cyber-Physical Systems Group, University of Konstanz, Konstanz, Germany

Supervised 10+ students in theses, projects, and seminars on embedded AI, automated machine learning, explainable AI, and machine learning for sensing.

Taught tutorials in undergraduate and graduate courses on operating systems and collective robotics.

Student Teaching Assistant Oct 2017 – Mar 2020
Institute of Mathematics and Image Computing, University of Lübeck, Lübeck, Germany

Led weekly tutorials and evaluated student coursework for undergraduate mathematics courses.

Proposals and Project Coordination

EU Horizon 2020 and German National Research Consortia Nov 2022 – Aug 2026
University of Konstanz, Konstanz, Germany

Wrote parts of research project proposals.

Supported the coordination of international research consortia, including project meetings, budget planning, and reporting to funding agencies.

Publications

First-Author Publications

Aust, T., T. Zhang, A. Reina, and H. Hamann (2026). "Bimodal Synchronization Performance: Why Noise and Sparse Connectivity Can Improve Collective Timing". In: *From Animals to Animats: The 18th International Conference on the Simulation of Adaptive Behavior*. DOI: 10.1007/978-3-032-39953-3_6.

Aust, T., E. Buss, F. Mohr, and H. Hamann (2025). "Automated Phytosensing: Ozone Exposure Classification Based on Plant Electrical Signals". In: *2025 IEEE Symposia on Computational Intelligence for Energy, Transport and Environmental Sustainability (CIETES)*. DOI: 10.1109/CIETES63869.2025.10995087.

Aust, T., C. K. Heck, E. Buss, and H. Hamann (2025). "Embedded Deep Learning for Bio-hybrid Plant Sensors to Detect Increased Heat and Ozone Levels". In: *Proceedings of IEEE SENSORS 2025*. DOI: 10.1109/SENSORS59705.2025.11330879.

Aust, T. (2024). "Classification and Forecasting of Physiological Data in Biological Systems". In: *Organic Computing*. DOI: 10.17170/kobra-202402269661.

Aust, T., E. Balta, A. Vatakis, and H. Hamann (2024). "Automatic classification of subjective time perception using multi-modal physiological data of air traffic controllers". In: *2024 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. DOI: 10.1109/SMC54092.2024.10831642.

Aust, T., M. S. Talamali, M. Dorigo, H. Hamann, and A. Reina (2022). "The Hidden Benefits of Limited Communication and Slow Sensing in Collective Monitoring of Dynamic Environments". In: *Proceedings of International Conference on Swarm Intelligence*. DOI: 10.1007/978-3-031-20176-9_19.

Co-Author Publications

Buss, E., T. **Aust**, and H. Hamann (2027). "Water stress detection from plant electrophysiology: A machine learning framework for irrigation management". In: *Computers and Electronics in Agriculture* 256, 112434. DOI: 10.1016/j.compag.2026.112434.

Hedberg, E., T. **Aust**, J. Sebastián-Azcona, V. Hernandez-Santana, G. Stemme, H. Hamann, A. D. Espejo, and N. Roxhed (2026). "Dynamic Phytohormone Trajectories Reveal Predictive Markers of Ozone Exposure Through Longitudinal Sampling and Machine Learning". In: *Plant, Cell & Environment*. DOI: 10.1111/pce.70967.

Buss, E., T. **Aust**, and H. Hamann (2025). "When Plants Respond: Electrophysiology and Machine Learning for Green Monitoring Systems". In: *Biomimetic and Biohybrid Systems. Living Machines 2025*. DOI: 10.1007/978-3-032-07448-5_21.

Buss, E., T. **Aust**, O. Hamburger, C. K. Heck, and H. Hamann (2025). "PhytoNode Upgraded: Energy-Efficient Long-Term Environmental Monitoring Using Phytosensing". In: *Advances in Information and Communication. FICC 2025*. DOI: 10.1007/978-3-031-84457-7_7.

Buss, E., T. **Aust**, M. Wahby, T.-L. Rabbel, S. Kernbach, and H. Hamann (2023). "Stimulus classification with electrical potential and impedance of living plants: Comparing discriminant analysis and deep-learning methods". In: *Bioinspiration & Biomimetics*. DOI: 10.1088/1748-3190/acbad2.

Hamann, H., T. **Aust**, and A. Reina (2020). "Guerrilla Performance Analysis for Robot Swarms: Degrees of Collaboration and Chains of Interference Events". In: *Proceedings of International Conference on Swarm Intelligence*. DOI: 10.1007/978-3-030-60376-2_11.

von Haxthausen, F., T. **Aust**, H. Schwegmann, S. Böttger, F. Ernst, V. García-Vázquez, M. Kleemann, and M. Kaschwich (2020). "Visual servoing for semi-automated 2D ultrasound scanning of peripheral arteries". In: *Proceedings on Automation in Medical Engineering*. DOI: 10.18416/AUTOMED.2020.

Awards and Certificates

Certificate "Good Academic Practice during Doctoral Studies" May 2025
University of Konstanz, Konstanz, Germany

1st Place, Kilometer 1 Idea Cup (1,000 Euro) Apr 2025
Kilometer 1, Konstanz, Germany

3rd Place, BR41N.IO Brain-Computer Interface Hackathon Oct 2024
g.tec medical engineering, at IEEE Int. Conf. on Systems, Man, and Cybernetics, Kuching, Malaysia

Best Bachelor Award Jan 2020
University of Lübeck, Lübeck, Germany

Technical Skills

Machine Learning: deep learning, transformers, time-series foundation models, time-series classification and forecasting, gradient boosting (XGBoost, LightGBM), automated machine learning, explainable AI, selective classification, uncertainty estimation, synthetic data generation
Signal Processing and Evaluation: denoising and filtering, feature extraction, anomaly detection, multimodal physiological signals, cross-subject validation, statistical hypothesis testing, calibration analysis, user studies
Robotics and Simulation: swarm and multi-robot systems, ARGoS, Kilobots and Kilogrid, ROS and Gazebo, KUKA LBR iiwa robot arm, visual servoing
Programming: Python, C, C++, MATLAB, SQL (MySQL), Bash
Frameworks: PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, ExecuTorch, Pandas, NumPy, SciPy, Matplotlib
Infrastructure: Linux, macOS, Git, Docker, Kubernetes, High Performance Computing clusters (SLURM)
Languages: German (native speaker), English (fluent)